

ID5040 - Engineering Plasticity - Homework #3

(Due date: 7-April-2025)

1. Consider the J_2 elastic plastic model with combined isotropic and kinematic hardening, whose governing equations are summarized below.

$$\text{Elastic constitutive law: } \dot{\sigma}_{ij} = 2\mu(\dot{\epsilon}_{ij} - \dot{\epsilon}_{ij}^p) + \bar{\lambda}\dot{\epsilon}_{kk}\delta_{ij},$$

$$\text{Yield criterion: } f(\underline{\beta}, s) := \sigma_{\text{eq}} - \sigma_y(s), \quad \sigma_{\text{eq}} := \sqrt{\frac{3}{2}\beta_{ij}\beta_{ij}}$$

$$\text{Flow rule: } \dot{\epsilon}_{ij}^p = \dot{\lambda} \frac{\partial f}{\partial \sigma_{ij}},$$

$$\text{Evolution of hardening variables: } \dot{\alpha}_{ij} = \frac{2}{3}K\dot{\epsilon}_{ij}^p, \quad \dot{s} = \sqrt{\frac{2}{3}\dot{\epsilon}_{ij}^p\dot{\epsilon}_{ij}^p}$$

$$\text{Kuhn-Tucker conditions: } f \leq 0, \dot{\lambda} \geq 0, \dot{\lambda}f = 0,$$

where α_{ij} denotes the backstress tensor, $\beta_{ij} := \sigma_{ij} - \alpha_{ij}$ is the effective stress tensor, and the positive constant K denotes the rate of kinematic hardening. Use the consistency condition to derive an expression for the consistency parameter $\dot{\lambda}$. Show that the elasto-plastic tangent modulus $\underline{\mathbf{C}}^t$ during plastic loading can be expressed in the form

$$C_{ijkl}^t = C_{ijkl} - \frac{9\mu^2}{\sigma_y^2} \frac{\beta_{ij}\beta_{kl}}{[3\mu + K + h(s)]}$$

where $h(s) = \frac{d\sigma_y}{ds}$ is the hardening modulus.

2. Consider homogeneous deformation of a body subjected to a history of deformation

$$[\underline{\epsilon}] = \begin{bmatrix} 1 & 0 & 0 \\ 0 & a & 0 \\ 0 & 0 & b \end{bmatrix} \epsilon$$

where a and b are constants and ϵ increases monotonically from 0 to 0.05. Using the radial return mapping algorithm, compute and plot the following as a function of ϵ for a power-law hardening material whose flow strength is given by the expression $\sigma_y(s) = \sigma_0(1 + 500s)^n$, where s is the equivalent plastic strain given by $\dot{s} = \sqrt{\frac{2}{3}\dot{\epsilon}_{ij}^p\dot{\epsilon}_{ij}^p}$.

(i) Stresses σ_{11}, σ_{22} and σ_{33} (all three curves in one plot).

(ii) Plastic strains $\epsilon_{11}^p, \epsilon_{22}^p$ and ϵ_{33}^p (all three curves in one plot).

Assume that Poisson's ratio $\nu = 0.3$ and $E/\sigma_0 = 500$ (choose any suitable value for E). Show results for two values of the hardening exponent, $n = 0$ (perfectly plastic) and $n = 0.1$, and two separate loading paths, $a = -1, b = 0$ (pure shear) and $a = b = -\frac{1}{2}$ (axisymmetric); i.e. eight plots in total.